Baselines First

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach every score with its row-level evidence.

Use complete fixtures, distinguish percentage points from relative changes and keep development separate from final evaluation.

Keep one habit: Require error-type counts and bounded conclusions as well as the total score.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary target: Fit a **majority label baseline** using **training** labels only

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Teach two 45-minute sessions. Fit a binary majority-label **baseline** from **training** labels and freeze its **prediction** before scoring. The invented set uses one frame per distinct trip; it is separate from Week 6 data.

Supplied candidate **prediction**s are comparison examples, not an implemented fitted model. Treat this as a first benchmark and **evidence** lesson, not a readiness test.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Related frames can share conditions and objects. Keeping the trip together prevents that trip from appearing in both development and evaluation sets.

Printed lesson and workbook

Six **training** label cards and four validation cards kept in order

Optional Python 3 editor with no extra packages

Blank comparison table with row reference **baseline** candidate and matches

Opening retrieval

Without opening the old lesson, explain one key idea from Dataset Cards. Give an example and a mistake that the idea helps you avoid.

Look for: Inspect row meaning missing values and repeated records

Reference example, not the only acceptable wording: One row is intended to represent one indoor practice trip.

The source is a teacher-authored fictional export for this lesson, duration uses minutes, and None means unknown rather than zero.

Without opening the old lesson, explain one key idea from Sampling and Intended Use. Give an example and a mistake that the idea helps you avoid.

Look for: Define the population and conditions in an intended-use statement

Reference example, not the only acceptable wording: Compare intended conditions with observed photo counts and distinct source plants. In Coverage v1, A has counts9,0,1,0 and B3,3,3,3 across day/A,day/B,evening/A,evening/B. Copying rows does not fill missing cells; coverage is not measured accuracy and unknown class/species selection remains a limit.

Without opening the old lesson, explain one key idea from Python and Table Readiness. Give an example and a mistake that the idea helps you avoid.

Look for: Specify a table schema

Reference example, not the only acceptable wording: activity: nonempty text; minutes: integer from 0 to 60 inclusive.

End goal

By the end, I can explain **baseline**s first using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Dataset Cards

Sampling and Intended Use

Python and Table Readiness

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A dataset contains 27 healthy and 3 spotted leaves. An always-healthy **baseline** gets what **accuracy** and spotted recall? What must a new model improve upon?

Concept to explain

A model must be compared with a simple transparent strategy on the same untouched examples.

Term	Meaning
baseline	simple comparison method
majority class	most frequent training label
rule baseline	transparent fixed decision rule
incremental value	measured gain beyond baseline

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach training-only baseline fitting and row-aligned scoring

> Keep training counts, evaluation scores and improvement units separate.

Complete source, labels and majority choice

A learner reports that a classifier gets 80 percent correct. Is that useful? We need to know the task, the examples and what a simple method achieves on those same examples. A **baseline** is a reference method.

This week our **baseline** ignores each scene and always predicts the label that is most common in the **training** set.

Use a new teacher-authored fictional dataset for predicting clear or blocked lanes. Each listed row describes one frame from a different trip on a familiar route; **training** and validation trips do not overlap. This is a new practice set, not the twelve-frame table from Week 8. All supplied labels have been resolved under the classroom rule. There are no review cases in this exercise, and no real navigation claim is made.

Set	Labels in row order	Use
Training	clear clear blocked clear blocked clear	Choose the constant label
Validation	blocked clear blocked clear	Score the fixed predictions
Final test	Not supplied in this exercise	Keep for later final evaluation

The **training** counts are four clear and two blocked. Our stated method therefore predicts clear for every new example. A **baseline** can still learn a value from **training** data; simple does not mean it uses no data.

If the two **training** counts tie, our rule chooses clear, declared before scoring.

Trace validation score and assumptions

The **output** is clear 2 4. The fixed **prediction** matches two of the four validation labels, so **accuracy** is 2/4 or 50 percent. The **training** majority share, 4/6, is not the validation **accuracy**.

The **baseline** does not inspect the validation labels to decide which constant to predict.

The code assumes nonempty **training** data with only clear and blocked labels. It counts matches on the supplied validation list; a percentage also requires a nonempty validation set.

Do not run it on empty **training** data and describe the default clear value as learned **evidence**. Other labels and unresolved review statuses need an explicit policy.

Term	Meaning
Baseline	A simple reference method used to judge a more complex approach
Majority label	The label with more examples than the other label in this binary task
Accuracy	Correct predictions divided by the number of scored examples
Percentage point	A unit for the difference between two percentages

Compare supplied corridor candidate

Suppose a candidate method supplies blocked, clear, clear, clear for the four validation rows in the table. These are supplied **prediction**s for a comparison exercise; we have not fitted that candidate.

It gets three of four correct, or 75 percent. The **baseline** gets two of four, or 50 percent. The candidate is ahead by one correct example and 25 **percentage point**s on this set.

Use identical row IDs, reference labels, exclusions and metric for both methods. Match **prediction**s to their rows even if files are reordered. A score on easier examples is not a fair comparison.

Neither this four-row result nor beating one **baseline** proves that the candidate will perform well on future trips.

> **The candidate misses the third validation row. Which rows does the baseline miss? Explain how the candidate changes the number correct. Why is the improvement 25 percentage points rather than a claim of 25 percent relative improvement?**

Record **training** counts, the fixed **prediction** and tie rule, validation IDs and labels, **prediction** lists and correct/total counts. Keep the final test separate from development decisions.

Next week we will search candidate rules using **training evidence**, then compare choices carefully.

Read separate pictured leaf facts

Make a simple **baseline** earn its place before claiming a better classifier.

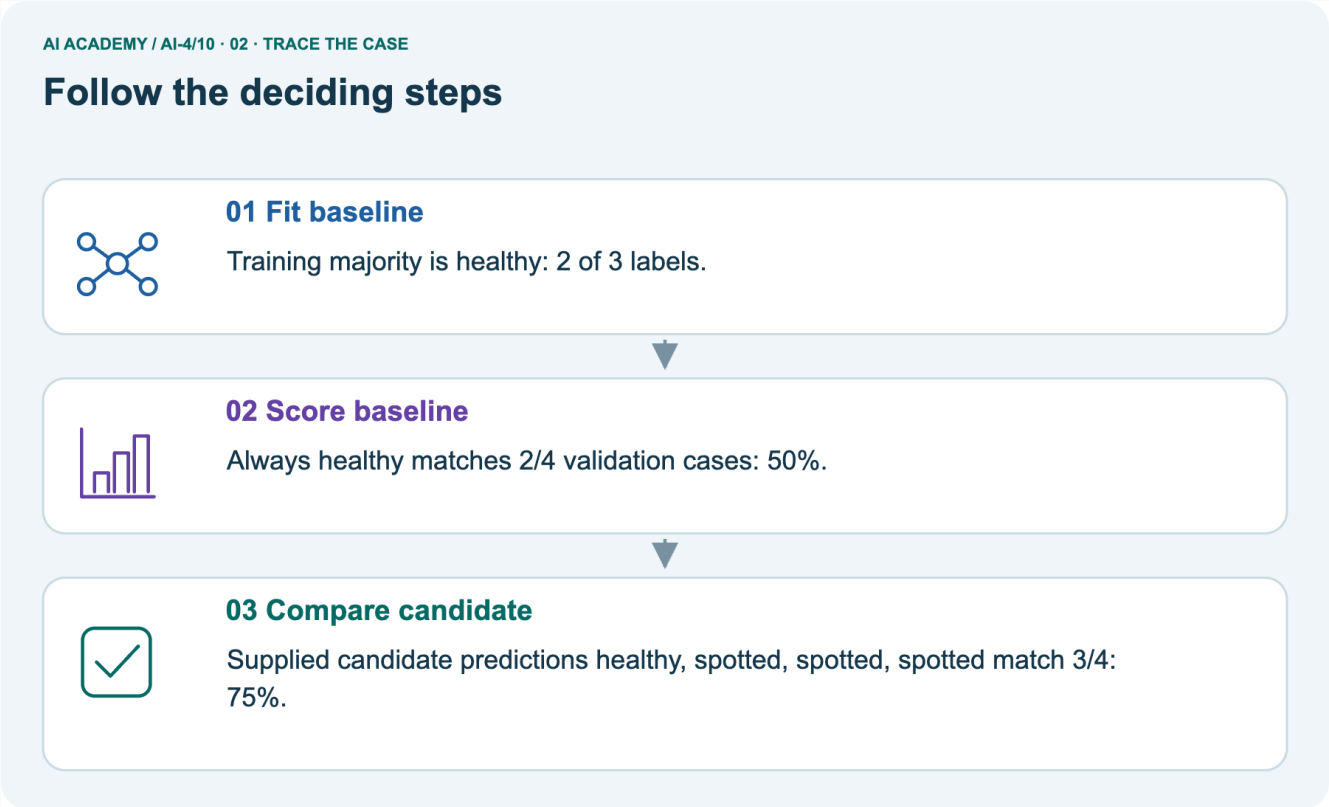
Training labels: healthy, healthy, spotted. Majority **baseline** predicts healthy. Validation references: healthy, spotted, healthy, spotted.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The pictured leaf example and corridor example both start at2/4 and compare a3/4 candidate, but their labels and wrong-row identities differ. Corridor baseline missesV1,V3; supplied candidate missesV3. Leaf baseline missesrows2,4; leaf candidate missesrow3.

Fixing only the leaf candidate's third prediction gives4/4 on that same development set; it is an answer-informed illustrative repair, not new independent validation. The original50%→75% difference is25percentage points; relative gain in accuracy is25/50=50%, a different quantity.

Trace picture and changed prediction



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Fit baseline	Training majority is healthy: 2 of 3 labels.
2. Score baseline	Always healthy matches 2/4 validation cases: 50%.
3. Compare candidate	Supplied candidate predictions healthy, spotted, spotted, spotted match 3/4: 75%.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

These supplied **prediction**s illustrate comparison. No actual model was trained, and four cases are limited **evidence**.

Change: Change the third candidate **prediction** from spotted to healthy.

Prediction: Candidate becomes correct on all 4 validation cases.

Reason: Only the previously wrong third decision changes; **baseline** stays fixed at 2/4.

▼ Additional imbalance/recall example

A test set has eighteen healthy and two spotted leaves. **Baseline** always predicts healthy.

Calculate and critique **accuracy**.

18/20=90%, but spotted recall is 0/2=0%. Compare with task-relevant metrics and the trained model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Build majority-class and one-**rule baseline**s, then calculate their results.

MODEL IDEA A model must be compared with a simple transparent strategy on the same untouched examples.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Test set has 12 healthy and 8 spotted leaves. Always-healthy **baseline** gets 12 ÷ 20 = **60% accuracy** but 0% spotted recall.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 Test set has 12 healthy and 8 spotted leaves.

STEP 2 Always-healthy **baseline** gets 12 ÷ 20 = **60% accuracy** but 0% spotted recall.

STEP 3 On the same 20 leaves, the visible-spot rule correctly labels 5 spotted leaves, misses 3 spotted leaves, correctly labels 10 healthy leaves and falsely marks 2 healthy leaves as spotted.

Accuracy=(5+10)/20=75%; spotted recall=5/8=62.5%.

STEP 4 A **learned model** must beat relevant **baseline** metrics—not only one convenient total.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

Exact explanation to model

Explain the added worked example

Key terms

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Baselines First

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Guide the calculation, then remove help

> Collect each learner's prediction before showing its key.

Current completion task and checked key

Completion task: List the four **baseline prediction**s and mark which validation rows are correct. Give correct/total and **accuracy**. Explain why 4/6 is not the answer.

Teacher check: **Prediction**s are clear, clear, clear, clear. Rows 2 and 4 match, giving 2/4 or 50 percent. The fraction 4/6 describes clear labels in **training**, not correct **prediction**s on the four validation rows.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for A **baseline** before a model. Proposed explanation: "A simple **baseline** does not need **training** data". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: This method learns its constant from **training** label counts, although it ignores **input** features.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

Can the candidate's **training accuracy** replace this validation comparison? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

No. Compare candidate and **baseline** on the same held-out cases and references; **training** fit answers a different question. Changed-case check: Candidate becomes correct on all 4 validation cases.

Only the previously wrong third decision changes; **baseline** stays fixed at 2/4.

Changed-case reasoning: Candidate becomes correct on all 4 validation cases. Only the previously wrong third decision changes; **baseline** stays fixed at 2/4.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: A test set has eighteen healthy and two spotted leaves. **Baseline** always predicts healthy.

Write the question to explain: Calculate and critique **accuracy**.

Reveal after **prediction**s: 18/20=90%, but spotted recall is 0/2=0%. Compare with task-relevant metrics and the trained model.

Use the earlier transfer question as additional practice. Reserve the fresh majority-fitting and candidate comparison task for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: the **baseline** misses rows 1 and 3; the candidate misses only row 3, adding one correct example. The difference is 75 minus 50 = 25 **percentage point**s.

Relative improvement in **accuracy** would be (75 minus 50)/50 = 50 percent; keep the measure explicit. Exit: equal **accuracy** does not establish identical mistakes or suitability.

Inspect **prediction**s by row, missed blocked cases, false blocked **prediction**s, sample coverage and consequences, then evaluate using an appropriate held out **process**.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the split roles and related-trip issue. Ask what a score means without a reference method.

Stage	Teacher action
7 to 18 minutes	Count training labels and write the constant before revealing validation labels. State the tie rule before any comparison.
18 to 32 minutes	Trace the code. Complete W1 and W2, separating training share from validation accuracy.
32 to 45 minutes	Build a row-by-row comparison for the supplied candidate. Complete the guided percentage-point question.
Session 2 start to 8 minutes	Retrieve what changes if the baseline constant is chosen using validation labels.
8 to 20 minutes	Discuss the two unfair comparisons and complete W3. Require common rows, labels and scoring policy.
20 to 35 minutes	Collect W4 independently before hints. Require the training counts, both prediction lists, scores and restrained claim.
35 to 45 minutes	Use W5 and the exit question to compare error patterns. Give the fresh retry after feedback and record support.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Keep predictions, observed or supplied outcomes and unmeasured cases separate.

Supplied inputs and checked guidance

Two-baseline lab V1–V4:majorityclear2/4,blocked recall0/2;predeclared marker>=6 rule3/4,blocked recall1/2. Source data invented,rule not fitted on validation. Probe tie→clear;empty→unavailable;changedvalidblocked,blocked,blocked,clear→constantclear1/4.

Independent trainingblocked3/clear2→blocked×4;validationclear,blocked,clear,blocked baseline2/4,candidateclear,blocked,blocked,blocked3/4,one extra correct/25percentage points;future readiness unsupported.

Predict, investigate and record

1. 1. Write the hypothesis and expected **evidence** before running.
2. 2. **Baseline** Challenge: calculate two **baseline**s on one fixed set, compare class-specific errors, and choose the honest benchmark.
3. 3. Run one normal, one boundary, one missing, and one shifted case.
4. 4. Calculate the metric from raw counts or row-level errors.
5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the experiment ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the current misconceptions

A simple **baseline** does not need **training** data: This method learns its constant from **training** label counts, although it ignores **input** features.

Training majority share is validation **accuracy**: Fit using **training** labels, then count matches separately on validation rows. Their label proportions can differ.

Beating a **baseline** proves the system is ready: It is one comparison on specified examples. Error types, coverage, uncertainty and final evaluation remain relevant.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve the fresh majority-fitting and candidate comparison task for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

MISCONCEPTION TO REPAIR Any trained model is more intelligent and therefore better than a simple rule.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected experiment transfer

> Keep this key until after a first attempt; record any help and prior exposure.

Selected independent task and checked key

Selected case: New **training** labels are blocked, clear, blocked, blocked, clear. New validation labels are clear, blocked, clear, blocked. Fit the same **baseline** rule and list its four **prediction**s.

A supplied candidate predicts clear, blocked, blocked, blocked. Score both on these rows and state the difference in correct examples and **percentage point**s. What claim remains unsupported?

Training counts are three blocked and two clear, so predict blocked four times. The **baseline** matches rows 2 and 4: 2/4 or 50 percent. The candidate matches rows 1,2,4: 3/4 or 75 percent.

It gains one correct example and 25 **percentage point**s. This tiny validation comparison does not prove future performance or readiness for use.

Judge trace, scope and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Choose from training evidence

Count clear and blocked in the worked **training** list. State the constant **prediction** and the declared tie rule. Explain why the validation majority does not choose this **baseline**.

Training has four clear and two blocked, so the constant is clear. A tie chooses clear under the declared classroom rule.

Validation labels score the stored **prediction**; using their majority to choose it would change the **fitting** procedure.

W2 Calculate the validation score

List the four **baseline prediction**s and mark which validation rows are correct. Give correct/total and **accuracy**. Explain why 4/6 is not the answer.

Predictions are clear, clear, clear, clear. Rows 2 and 4 match, giving 2/4 or 50 percent. The fraction 4/6 describes clear labels in **training**, not correct **prediction**s on the four validation rows.

W3 Make the comparison fair

A candidate is scored on ten easy rows and the **baseline** on four difficult rows. Describe a fair comparison and two details you must record.

Score both methods on the same intended validation examples with matched IDs, reference labels, metric and exclusion policy. Record **prediction**s, correct/total counts and the **training**-derived **baseline** rule.

Different example sets do not isolate a method difference.

W5 Explain the hidden error

A **training**-derived clear **baseline** gets nine of ten validation cases correct; the labels are nine clear and one blocked. State its **accuracy** and missed case. If a candidate also gets nine correct, must it miss the same case?

Explain.

Accuracy is 90 percent, and the **baseline** misses the sole blocked case. A candidate could also score 90 percent by correctly identifying blocked but missing one clear case.

Equal totals do not imply identical errors; inspect **prediction**s by row and error type.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

18/20=90%, but spotted recall is 0/2=0%. Compare with task-relevant metrics and the trained model.

Additional transfer practice question

A dataset has 18 grass and 2 fern examples. A **baseline** always predicts grass. Compute **accuracy** and name a serious limitation.

Reasoned practice answer

Accuracy is 18/20 = 90%, but fern recall is 0/2. The high total hides complete failure on the minority class.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A 90% **baseline** necessarily serves both classes well.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Accuracy=27/30=90%; spotted recall=0/3=0%. A new model should be compared on task-relevant metrics and error consequences, not only **accuracy**.

Fit a majority **baseline** using **training** labels, without using final labels to choose the class.

Explanations for the additional workbook examples

W1 Choose from training evidence

W2 Calculate the validation score

W3 Make the comparison fair

W4 Fit and compare independently

New **training** labels are blocked, clear, blocked, blocked, clear. New validation labels are clear, blocked, clear, blocked. Fit the same **baseline** rule and list its four **prediction**s.

A supplied candidate predicts clear, blocked, blocked, blocked. Score both on these rows and state the difference in correct examples and **percentage point**s. What claim remains unsupported?

Training counts are three blocked and two clear, so predict blocked four times. The **baseline** matches rows 2 and 4: 2/4 or 50 percent. The candidate matches rows 1,2,4: 3/4 or 75 percent.

It gains one correct example and 25 **percentage point**s. This tiny validation comparison does not prove future performance or readiness for use.

W5 Explain the hidden error

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Record majority and transparent-**rule baseline**s for the Plant Classifier using the frozen test plan.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a **baseline**, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Use labelled cards and matching ticks before fractions. Supply a blank row-aligned comparison table without **prediction**s.

Extension: construct two different **prediction** lists with identical **accuracy** but different missed rows, and explain how to report both.

Require W4 blocked from counts 3 versus 2, **baseline** 2/4, candidate 3/4, one extra correct and 25 **percentage point**s. Fresh retry: **training** clear, blocked, blocked, clear is tied, so choose clear by the declared rule.

Validation blocked, clear, blocked, blocked gives **baseline** 1/4 or 25 percent. Candidate blocked, clear, clear, blocked gives 3/4 or 75 percent: two extra correct and 50 **percentage point**s.

Require the tie rule and explain why validation majority blocked cannot replace the stored constant.

Delayed return

Suggested return points in this level: ai-4/11, ai-4/13, ai-4/16. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 11: Decision Trees. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week11: Decision Trees. Carry forward the **training**-derived **baseline**, declared tie policy and row-aligned validation comparison. Primary reference: https://scikit-learn.org/stable/modules/generated/sklearn.dummy.DummyClassifier.html . This classroom exercise declares its own clear-on-tie policy; do not assume identical tie behavior from every library.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching target: Fit a **majority label baseline** using **training** labels only

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: List the four **baseline prediction**s and mark which validation rows are correct. Give correct/total and **accuracy**. Explain why 4/6 is not the answer.

Reasoned practice answer: **Prediction**s are clear, clear, clear, clear. Rows 2 and 4 match, giving 2/4 or 50 percent. The fraction 4/6 describes clear labels in **training**, not correct **prediction**s on the four validation rows.

Check every learner before discussion: Use the worked case for A **baseline** before a model. Proposed explanation: “A simple **baseline** does not need **training** data”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: This method learns its constant from **training** label counts, although it ignores **input** features.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change the third candidate **prediction** from spotted to healthy. Require a **prediction** and a reason before comparing. Expected result: Candidate becomes correct on all 4 validation cases.

Reason: Only the previously wrong third decision changes; **baseline** stays fixed at 2/4.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Earlier wording, alternatives and picture transcripts**

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

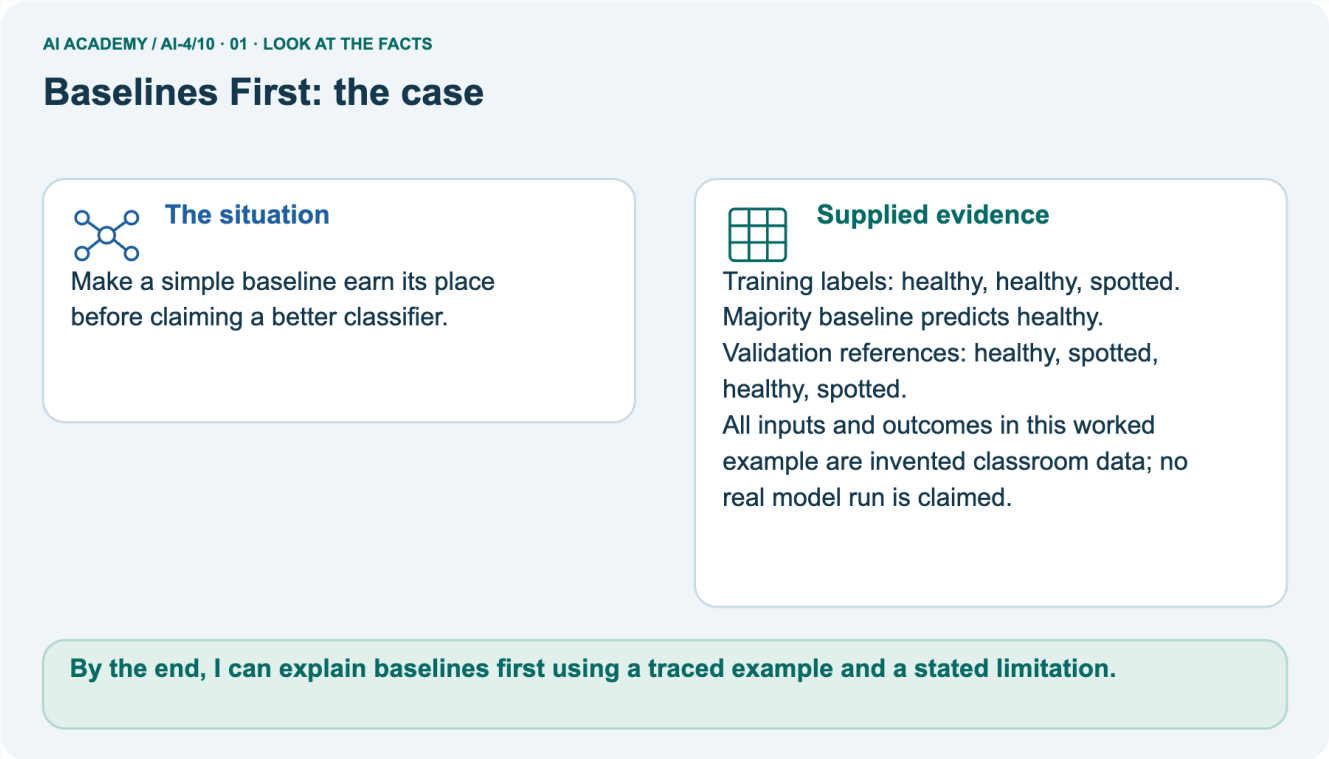
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Baselines First: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

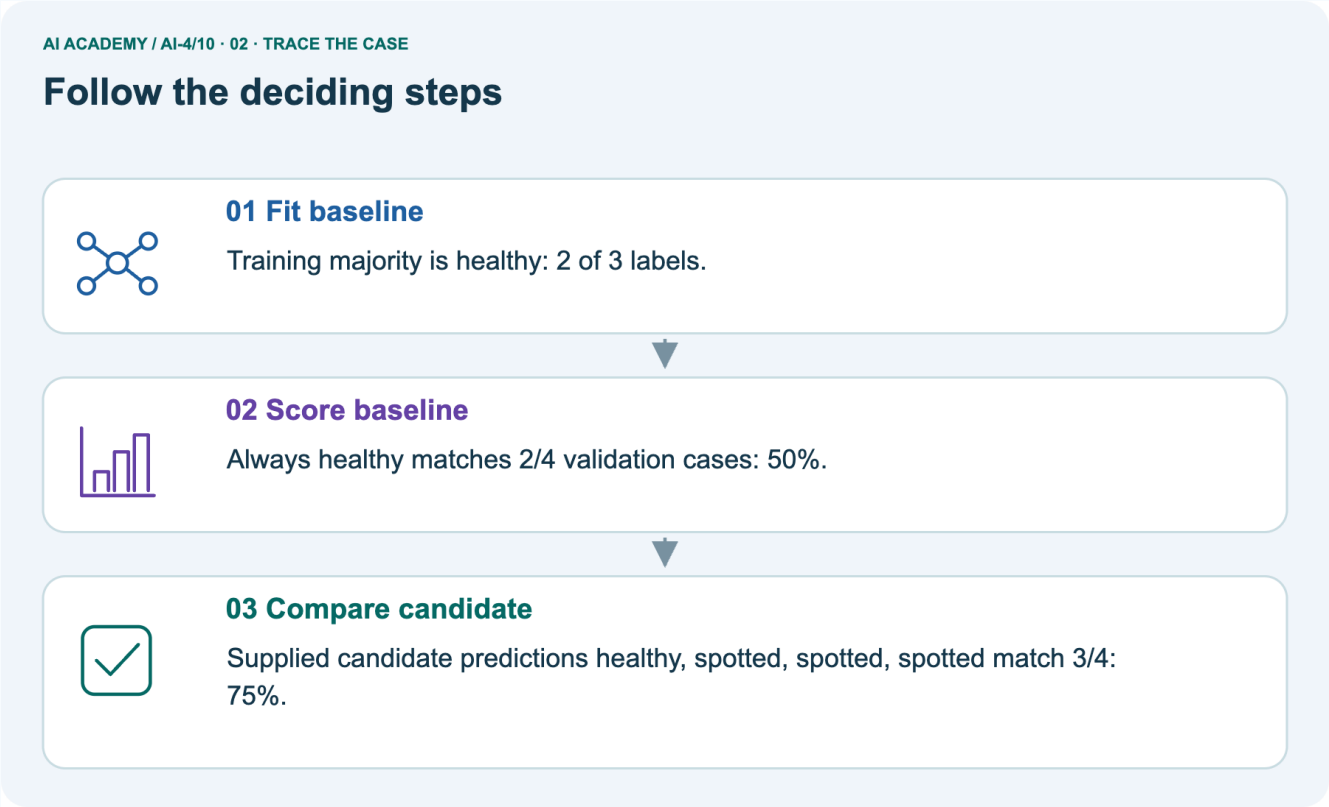
Make a simple **baseline** earn its place before claiming a better classifier. Supplied facts: **Training** labels: healthy, healthy, spotted. Majority **baseline** predicts healthy.

Validation references: healthy, spotted, healthy, spotted. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



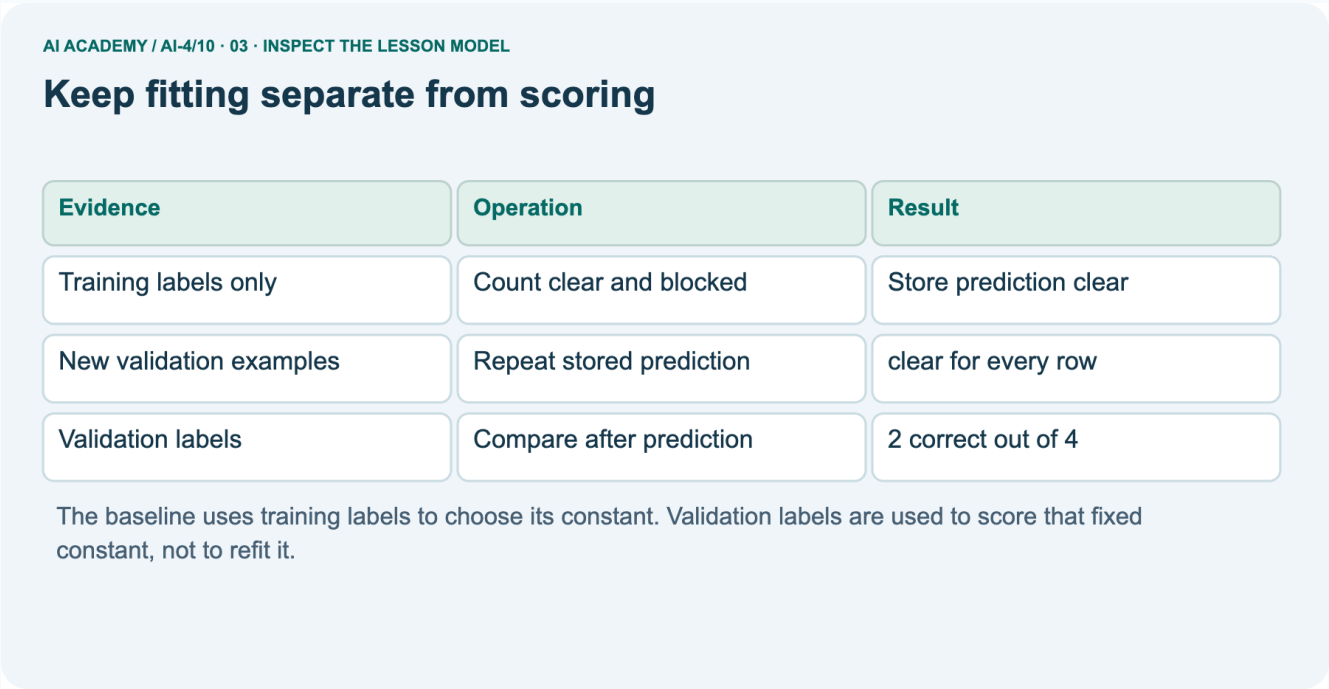
Follow the labelled picture with the matching explanation. Open the vector diagram

Fit **baseline**: **Training** majority is healthy: 2 of 3 labels. → Score **baseline**: Always healthy matches 2/4 validation cases: 50%. → Compare candidate: Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Keep fitting separate from scoring



Follow the labelled picture with the matching explanation. Open the vector diagram

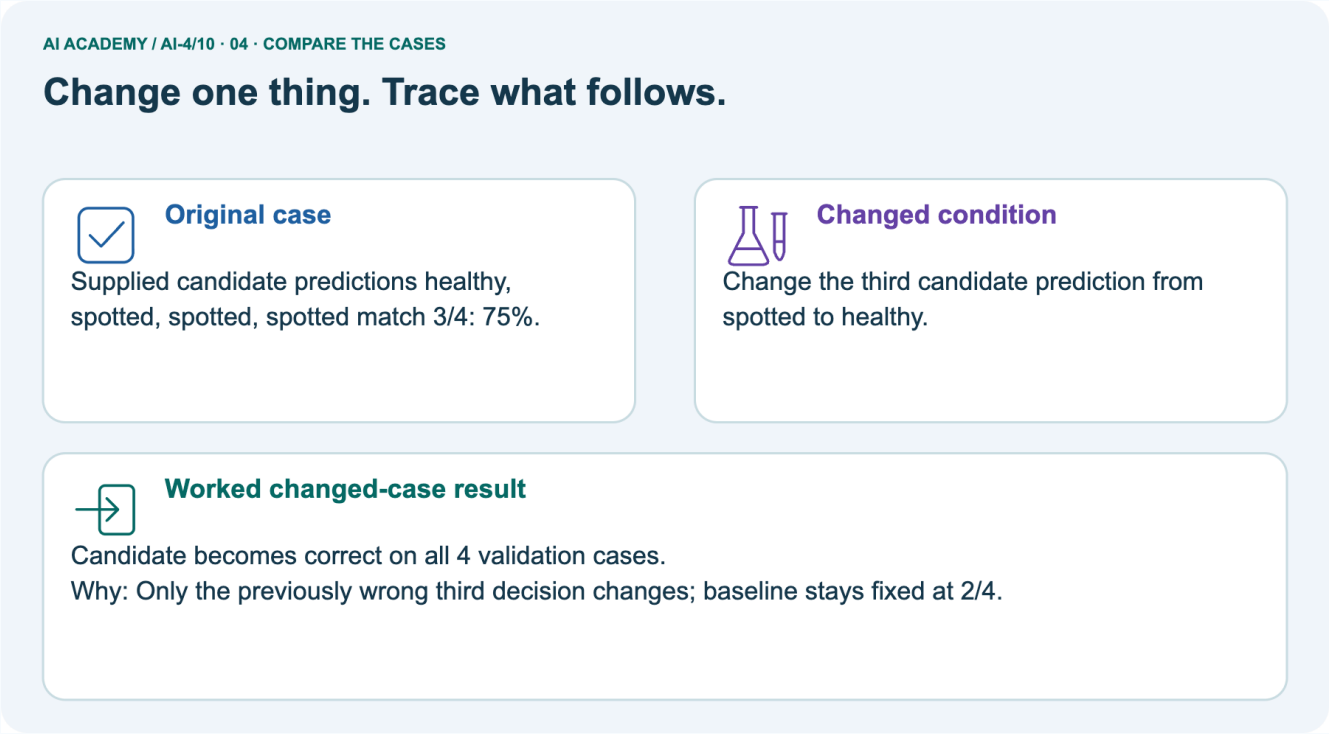
Keep **fitting** separate from scoring. **Training** labels only | Count clear and blocked | Store **prediction** clear. New validation examples | Repeat stored **prediction** | clear for every row.

Validation labels | Compare after **prediction** | 2 correct out of 4

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%. Changed condition: Change the third candidate **prediction** from spotted to healthy.

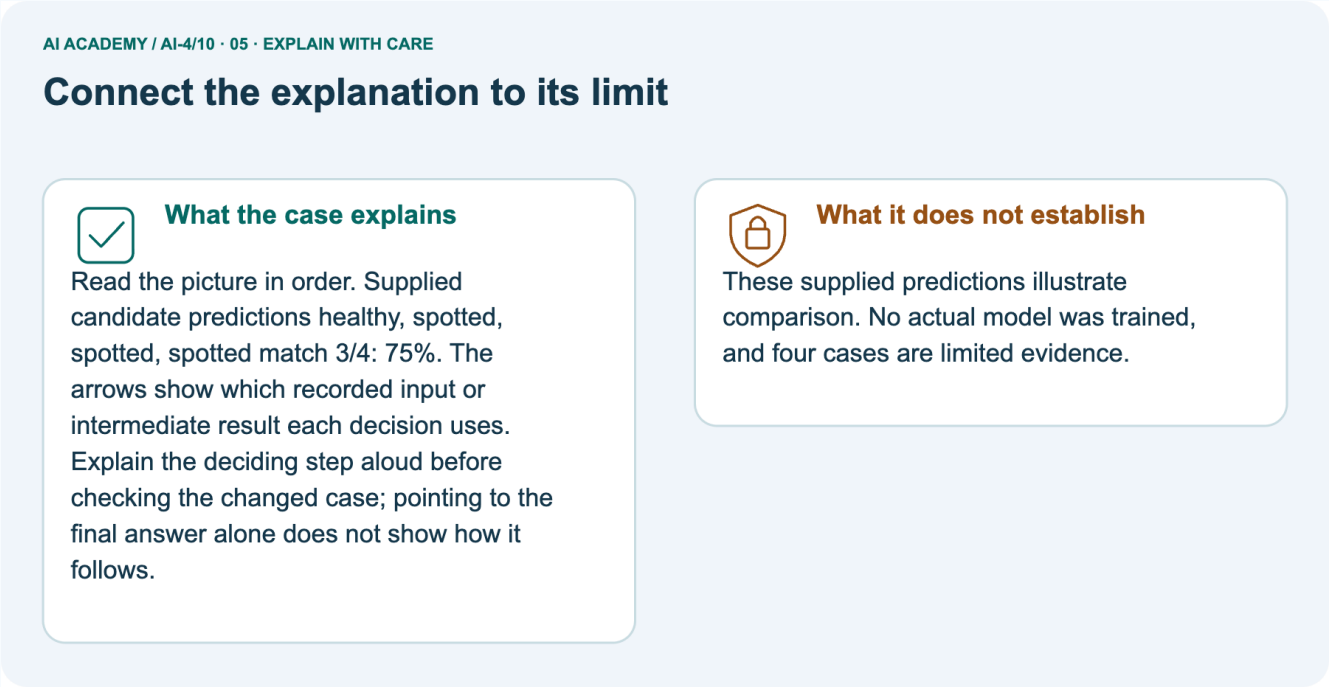
Expected result: Candidate becomes correct on all 4 validation cases. Reason: Only the previously wrong third decision changes; **baseline** stays fixed at 2/4.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Only the previously wrong third decision changes; **baseline** stays fixed at 2/4.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Supplied candidate **prediction**s healthy, spotted, spotted, spotted match 3/4: 75%. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: These supplied **prediction**s illustrate comparison.

No actual model was trained, and four cases are limited **evidence**.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: These supplied **prediction**s illustrate comparison. No actual model was trained, and four cases are limited **evidence**.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Can the candidate's **training accuracy** replace this validation comparison? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. Compare candidate and **baseline** on the same held-out cases and references; **training** fit answers a different question. Changed-case check: Candidate becomes correct on all 4 validation cases.

Only the previously wrong third decision changes; **baseline** stays fixed at 2/4.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Baselines First: the case

02. Follow the deciding steps

03. Keep fitting separate from scoring

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.